

Session- Variant Calling and Consensus Generation

BU-ISCIII

Unidades Comunes Científico Técnicas - SGSAFI-ISCIII

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Mapping against reference genome and Variant Calling :

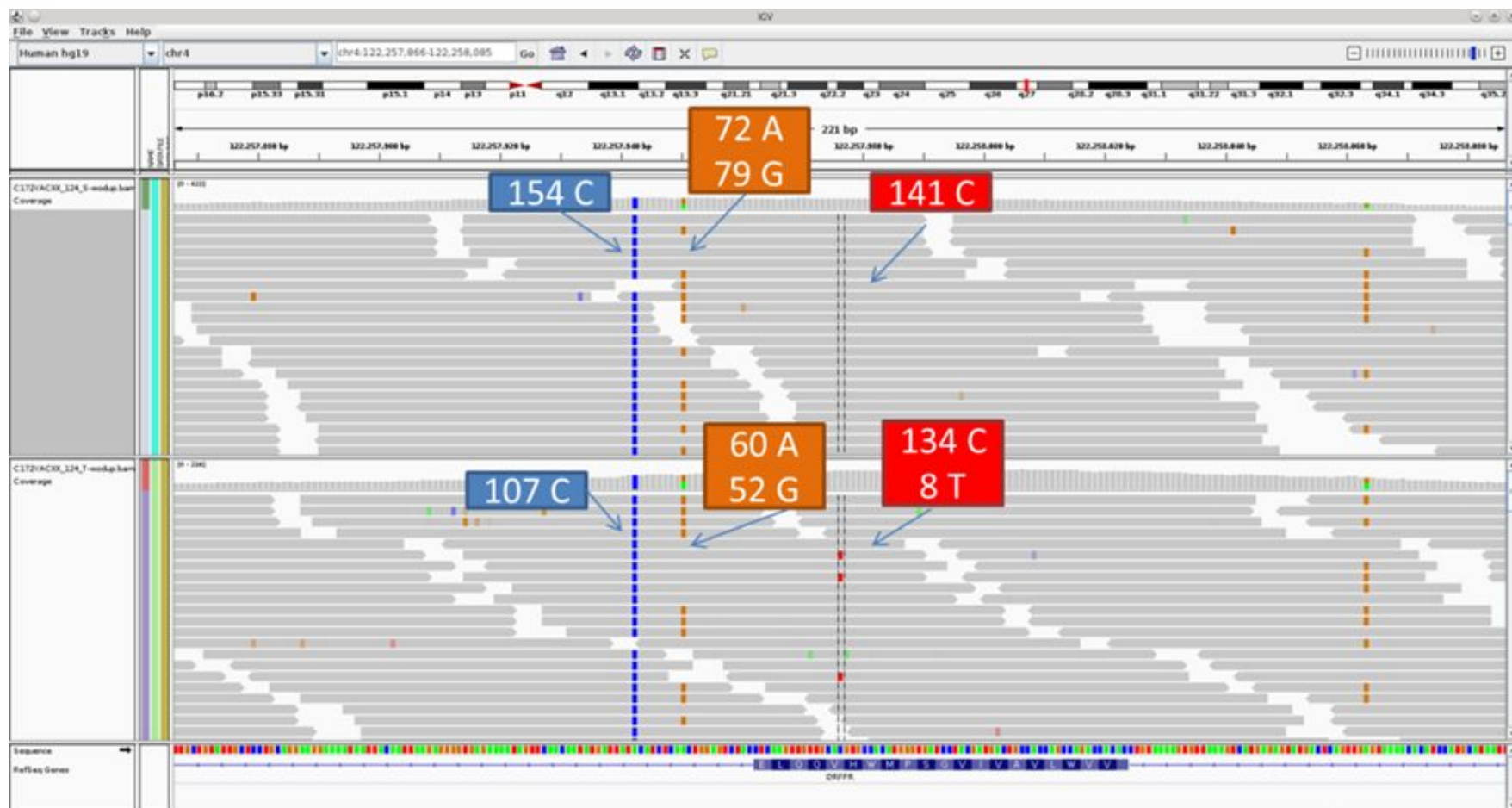
- Variant Calling
- Source of error and mitigation strategies
- VCF and bed format
- IVAR, LOFREQ, VARSCAN2
- Consensus generation: approximations

Variant Calling

- Variant calling concept is simple:

Find positions in our reads different from the reference.

- We start with our sequences mapped against our reference genome, and we walk through every column of the alignment counting the number of alleles found and comparing them against the reference.



Sources of error and mitigation strategies



Sample processing

Polymerase error



Sequencing

Polymerase error
Sequencing chemistry
Reaction detection.
Base calling



Read Mapping

Genome duplication
Structural variants



SNP Calling

Base Quality scores
Mapping quality scores
Filtering thresholds

Adapted from Olson et al. Frontiers in Genetics. 2015

Sources of error and mitigation strategies

- **Sample processing errors.**
 - Random errors.
 - Associated with polymerase errors . (1 in 10^{2-3} bases)
 - Homopolymers and tandem repeats experience higher indel error rates.
- **Solutions:**
 - Paired-end libraries.
 - Minimization of PCR cycles.

Adapted from Olson et al. Frontiers in Genetics. 2015

Sources of error and mitigation strategies

- **Sequencing:**
 - Dependent on the platform.
 - Can be random and systematic.
 - 6% Illumina, 50% Roche (Ross et al.2013)
 - P.e Illumina commits error in the G/T channels.
- **Solutions:**
 - Strand bias.

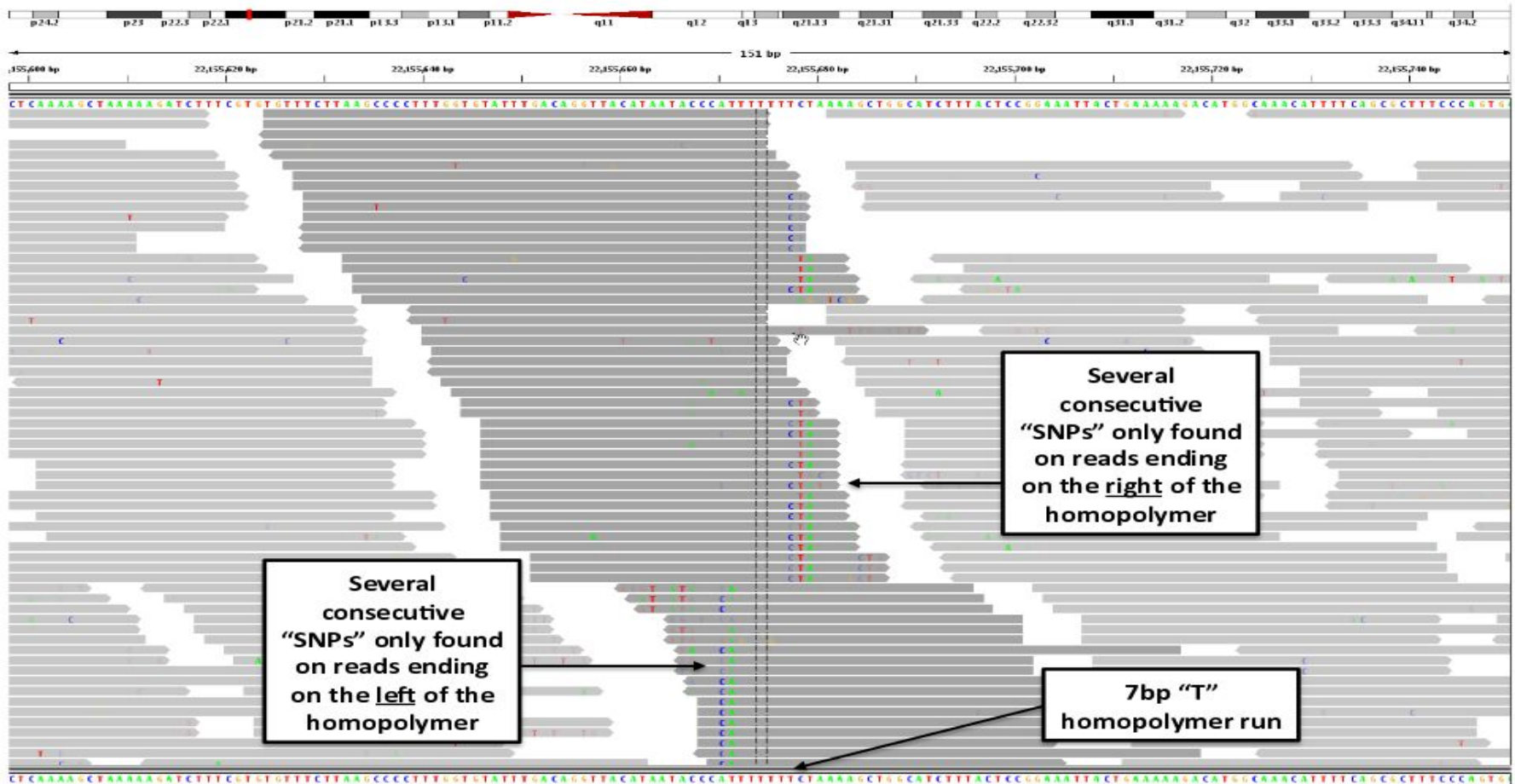
Adapted from Olson et al. Frontiers in Genetics. 2015

Sources of error and mitigation strategies

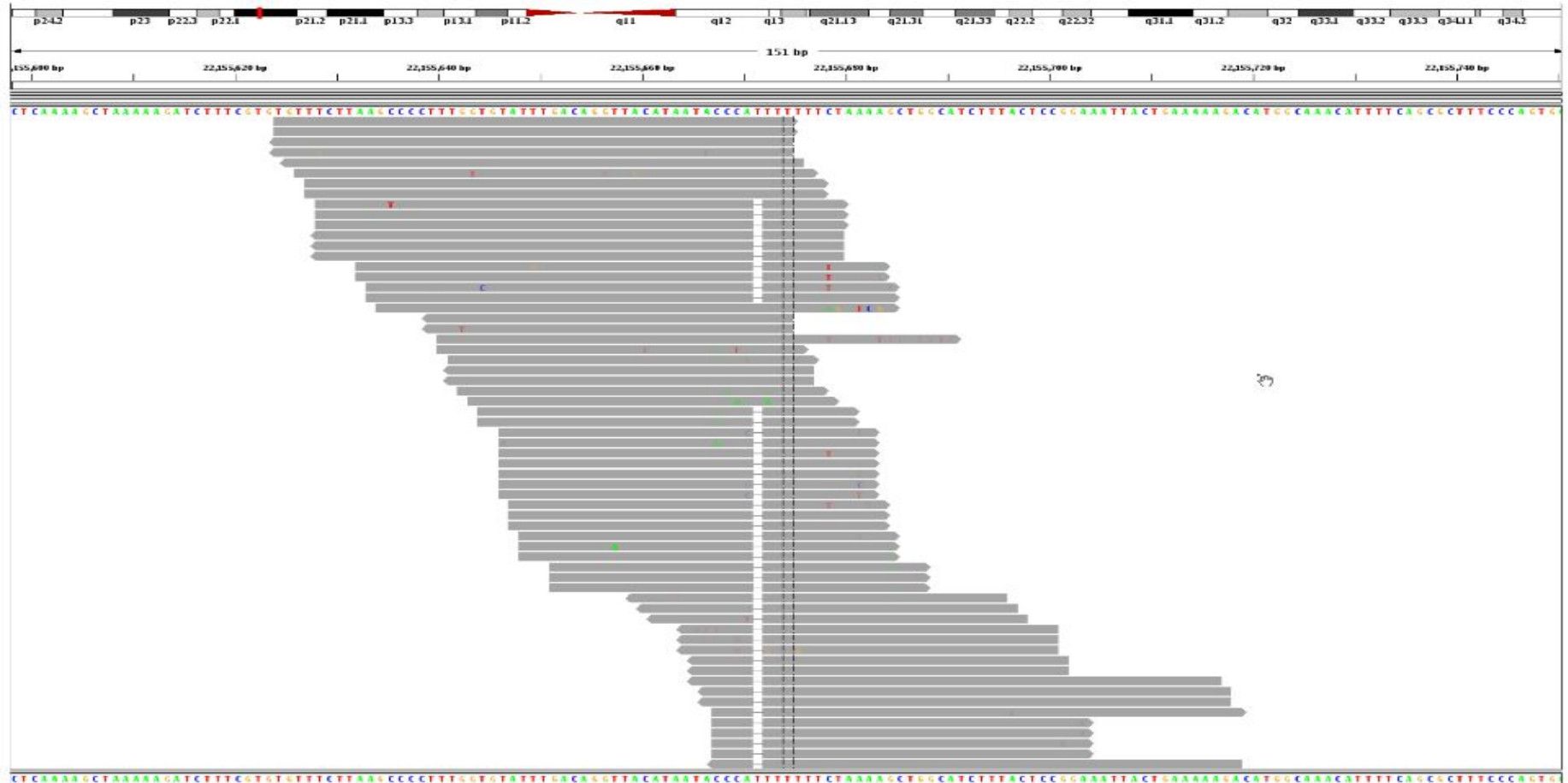
- **Mapping errors:**
 - Genomic duplication and structural variation.
 - High diverse areas.
- **Solutions**
 - Paired-end libraries.
 - Long reads / fragments.
 - MAPQ
 - Realignment around indels.

Adapted from Olson et al. Frontiers in Genetics. 2015

Sources of error and mitigation strategies



Sources of error and mitigation strategies



Sources of error and mitigation strategies

- **SNP calling step**
 - Errors may result in base calling errors.
 - FP and FN calls.
- **Solutions**
 - Strand bias
 - Base quality rank sum
 - MAPQ
 - Hard filters:
 - Depth of coverage
 - Minimum base call frequency.

Adapted from Olson et al. Frontiers in Genetics. 2015

Reference selection

- Critical step <- Bias which SNPs are called.
- SNPs in genes not present in the reference WON'T be called.
- Less effect in clonal bacteria.
- Number of SNPs called vary **A LOT!**
- **Solutions:**
 - Kmerfinder

Repetitive/Phage regions filtering

- **PHASTER**
- We can remove/mask phage/repetitive regions where reads won't map.
- This way those areas will be out of analysis.
- Problem: those areas could be important!

VCF format

VCF header

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

Mandatory header lines

Optional header lines (meta-data about the annotations in the VCF body)

Body

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13	0/0:29
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0/1:100	2/2:70
1	5	.	A	G	.	PASS	.	GT:GQ	1/0:77	1/1:95
1	100	.	T		.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3	0/0:20

Reference alleles (GT=0)

Alternate alleles (GT>0 is an index to the ALT column)

Deletion

SNP

Large SV

Insertion

Other event

Phased data (G and C above are on the same chromosome)

Bed format

chromosome	start	end	score	name	strand	thickstart	thickend	RGB
chr7	127471196	127472363	Pos1	0	+	127471196	127472363	255,0,0
chr7	127472363	127473530	Pos2	0	+	127472363	127473530	255,0,0
chr7	127473530	127474697	Pos3	0	+	127473530	127474697	255,0,0
chr7	127474697	127475864	Pos4	0	+	127474697	127475864	255,0,0
chr7	127475864	127477031	Neg1	0	-	127475864	127477031	0,0,255
chr7	127477031	127478198	Neg2	0	-	127477031	127478198	0,0,255
chr7	127478198	127479365	Neg3	0	-	127478198	127479365	0,0,255
chr7	127479365	127480532	Pos5	0	+	127479365	127480532	255,0,0
chr7	127480532	127481699	Neg4	0	-	127480532	127481699	0,0,255

OBLIGATORIOS

OPCIONALES

Mpileup format

Sequence	Position	Reference Base	Read Count	Read Results	Quality
seq1	272	T	24	,.\$. . . . , , , , , , , , , , , , ^+.	<<<+>;<<<<<<<<<<=<;<;7<&
seq1	273	T	23	, , , , , , , , , , , , A	<<<;<<<<<<<<<3<=<<<;<<+
seq1	274	T	23	,. \$. . . . , , , , , , , , , , , .	7<7>;<;<<<<<<<<=<;<;<<6
seq1	275	A	23	, \$. . . . , , , , , , , , , , , ^1.	<+>;9*<<<<<<<<=<<:>;<<<<
seq1	276	G	22	. . . T, , , , , , , , , , .	33>;+<<7=7<<7<&<<1>;<<6<
seq1	277	T	22	. . . , , , , . C. , , , , . G.	+7<;<<<<<<&<=<<:>;<<&<
seq1	278	G	23	. . . , , , , , , , , , . ^k.	%38*<<;<7<<7<=<<<;<<<<
seq1	279	C	23	A..T, , , , , , , , , .	75&<<<<<<<<=<<<9<<:<<<

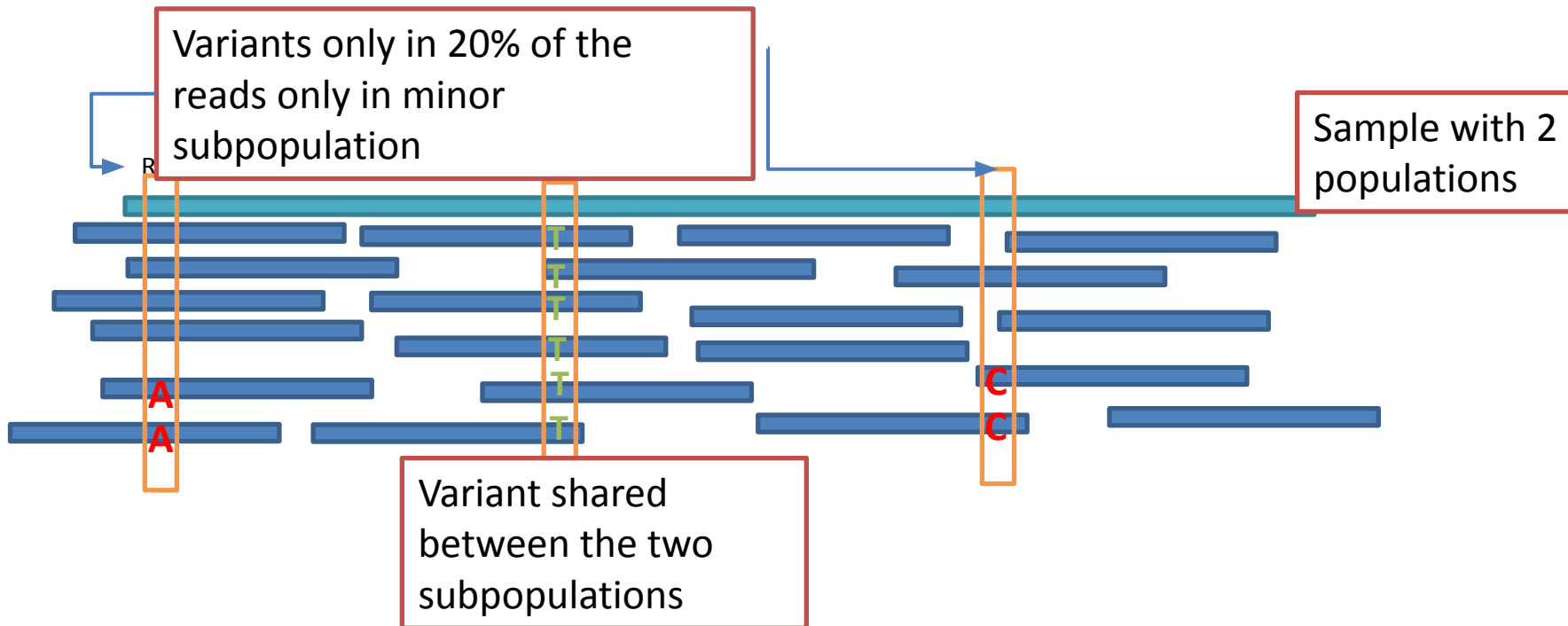
Mpileup format

Column 5: The bases string [\[edit\]](#)

- . (dot) means a base that matched the reference on the forward strand
- , (comma) means a base that matched the reference on the reverse strand
- </> (less-/greater-than sign) denotes a reference skip. This occurs, for example, if a base in the reference genome is intronic and a read maps to two flanking exons. If quality scores are given in a [sixth column](#), they refer to the quality of the read and not the specific base.
- AGTCN (upper case) denotes a base that did not match the reference on the forward strand
- agtcn (lower case) denotes a base that did not match the reference on the reverse strand
- A sequence matching the [regular expression](#) `\+[0-9]+\[ACGTNacgtn\]+` denotes an insertion of one or more bases starting from the next position. For example, `+2AG` means insertion of AG in the forward strand
- A sequence matching the regular expression `\-[0-9]+\[ACGTNacgtn\]+` denotes a deletion of one or more bases starting from the next position. For example, `-2ct` means deletion of CT in the reverse strand
- ^ (caret) marks the start of a read segment and the ASCII of the character following '^' minus 33 gives the mapping quality
- \$ (dollar) marks the end of a read segment
- * (asterisk) is a placeholder for a deleted base in a multiple basepair deletion that was mentioned in a previous line by the `\-[0-9]+\[ACGTNacgtn\]+` notation

Viral subpopulation - Quasispecies

- Just as in clonal subpopulations in tumor samples, we can have viral subpopulations called quasispecies in viral samples.
- We detect them using the alternative allele frequency.



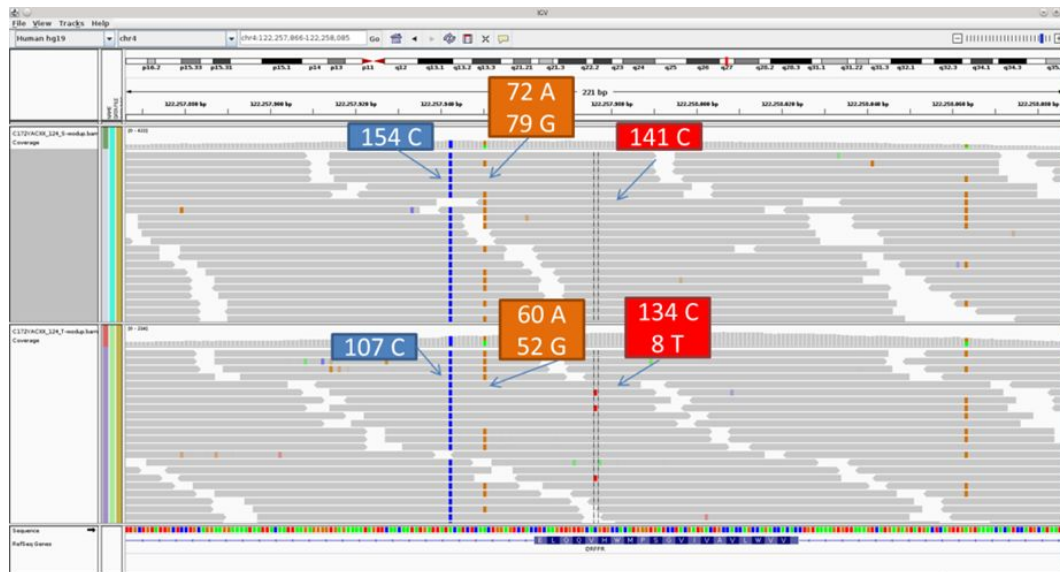
Population Allele frequency vs Sample Allele frequency

- **Population allele frequency**: probability of finding an allele in the population. Number of individuals carrying an allele vs total of individuals in the population.



Population Allele frequency vs Sample Allele frequency

- **Alternate/Base allele frequency**: number of reads supporting the alternate allele vs total of reads.



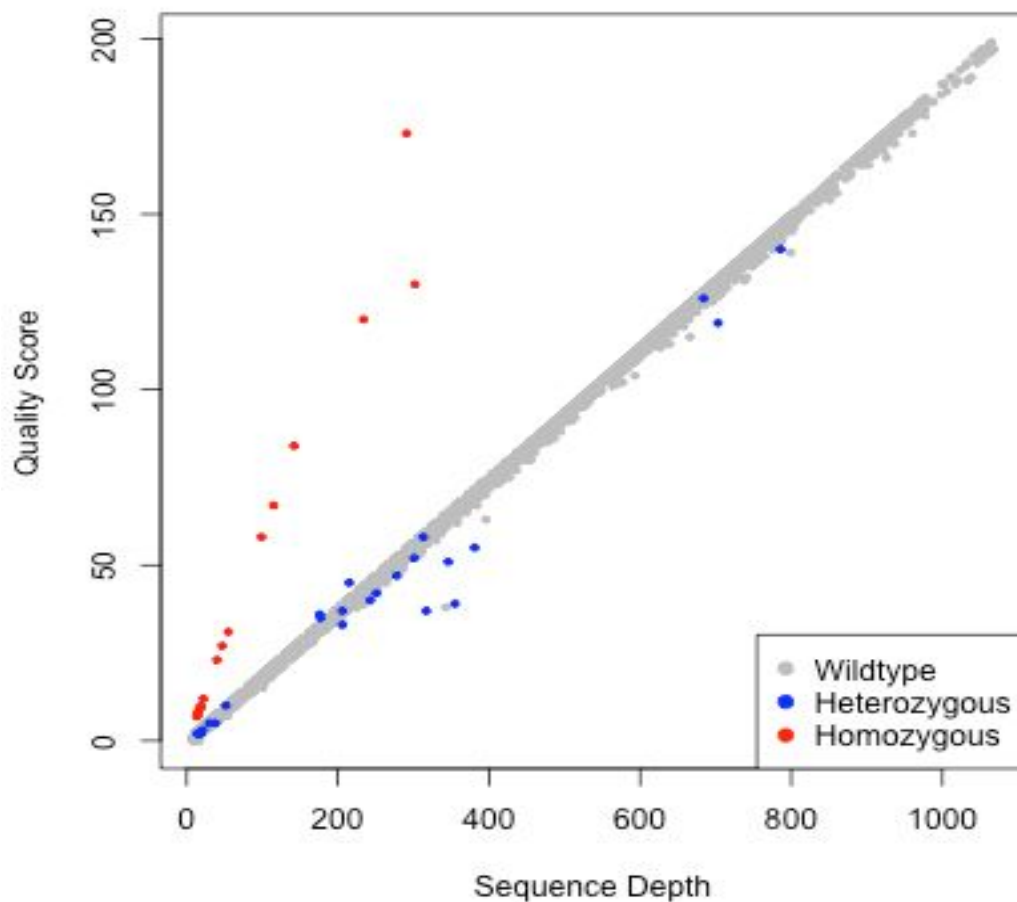
VARSCAN2

- Uses a heuristic/statistic method instead of bayesian.
- Allows more flexibility and hard filters.

```
samtools pileup -f reference.fasta myData.bam | java -jar VarScan.v2.2.jar pileup2snp
```

<http://varscan.sourceforge.net/>

VARSCAN2



IVAR

primalscheme primer panels for multiplex PCR



Design a new scheme

Reset defaults



FASTA file...

Select a file...

Options



High-GC mode



Pinned

i Use the [standard protocol](#) for these settings.

Amplicon size

400

Min/max will be set at 5% either side of target.

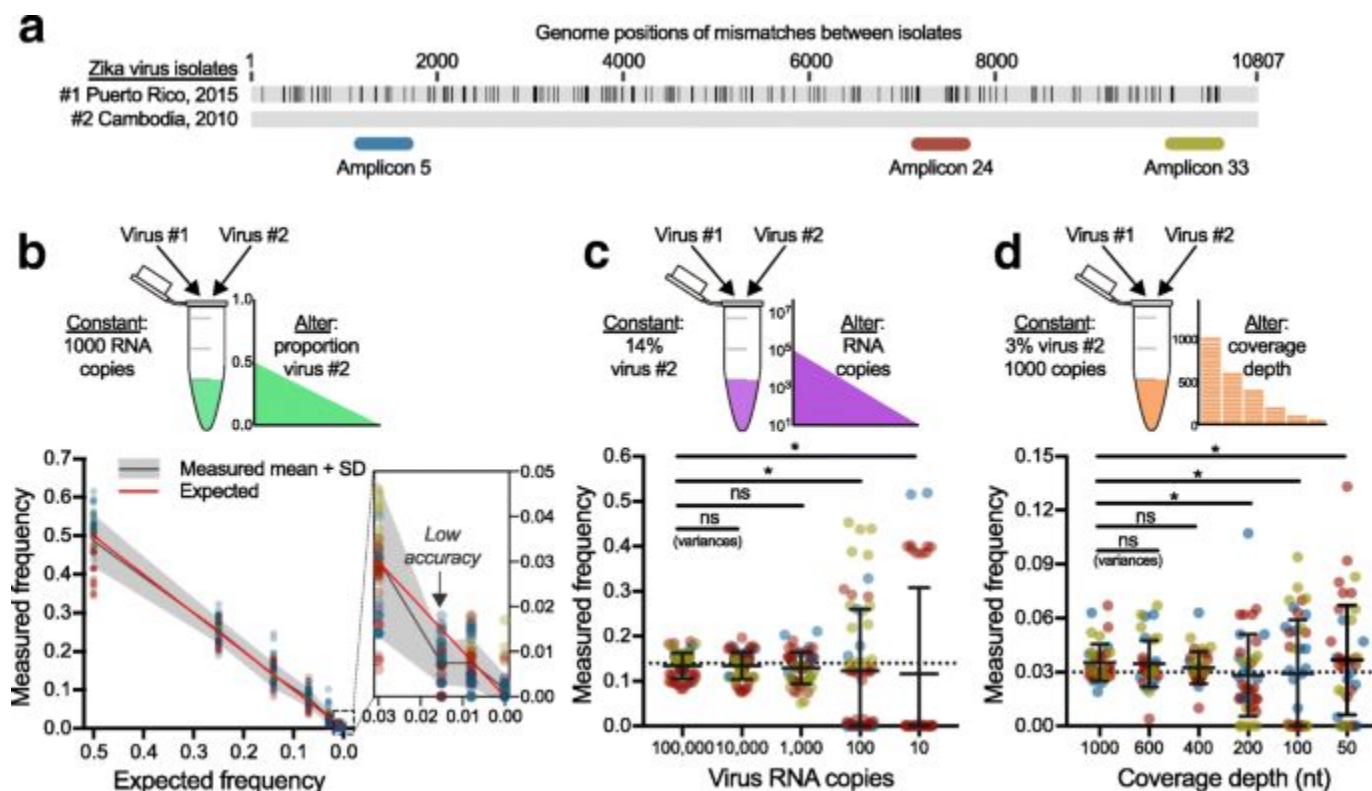
Scheme name

e.g., nCov-400

A short name/prefix for your scheme, no spaces.

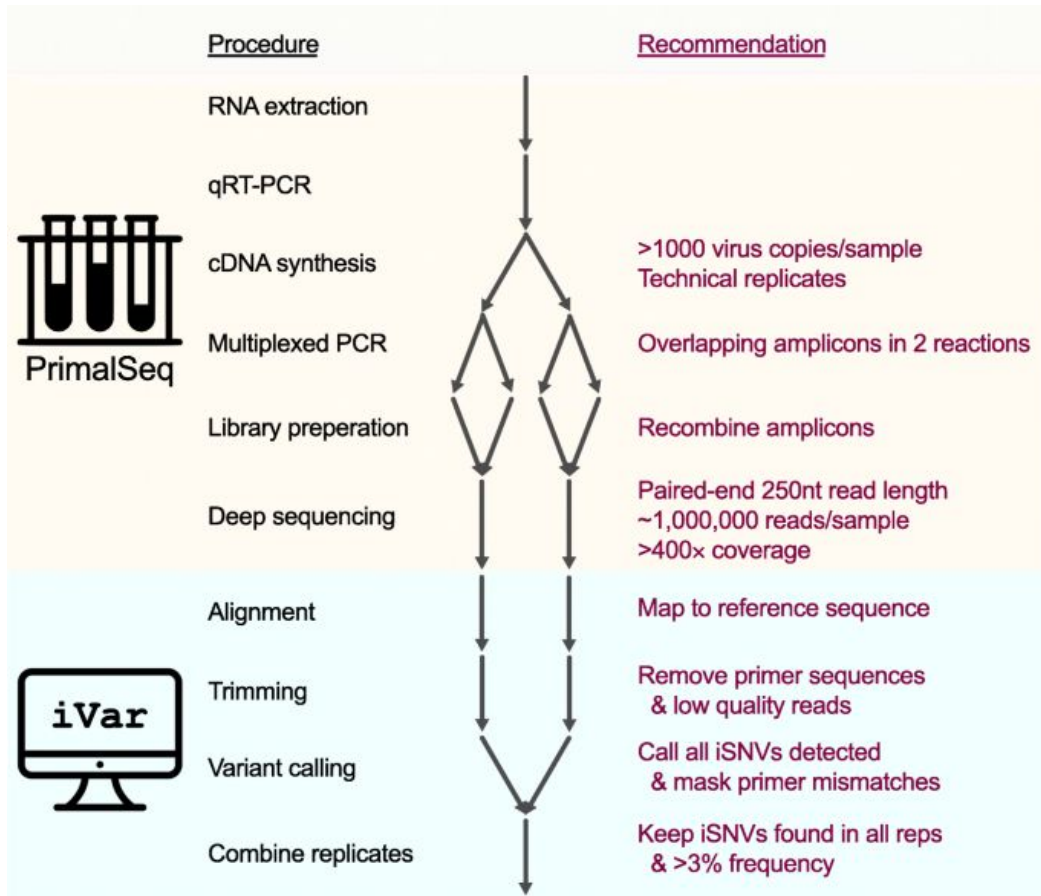
Quick et al. Nature Protoc. 2017

IVAR



Grubaugh et al. Genome Biology. 2019

IVAR



Grubaugh et al. Genome Biology. 2019

IVAR

Command	Description
trim	Trim reads in aligned BAM
variants	Call variants from aligned BAM file
filtervariants	Filter variants across replicates or multiple samples aligned using the same reference
consensus	Call consensus from aligned BAM file
getmasked	Detect primer mismatches and get primer indices for the amplicon to be masked
removereads	Remove reads from trimmed BAM file
version	Show version information
trimadapter	(EXPERIMENTAL) Trim adapter sequences from reads

Grubaugh et al. Genome Biology. 2019

IVAR

- Input Options Description

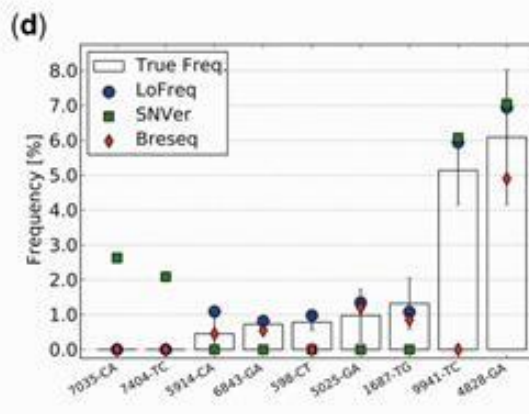
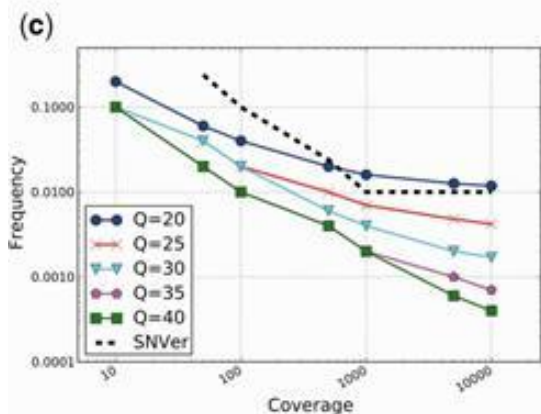
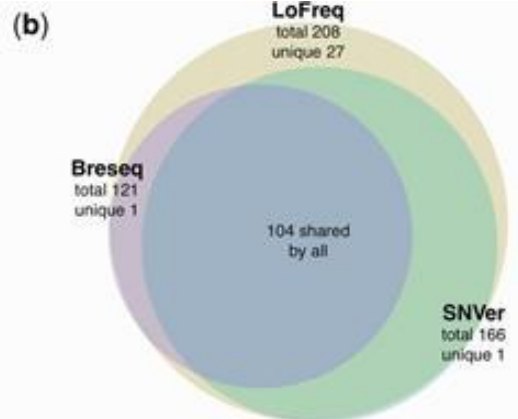
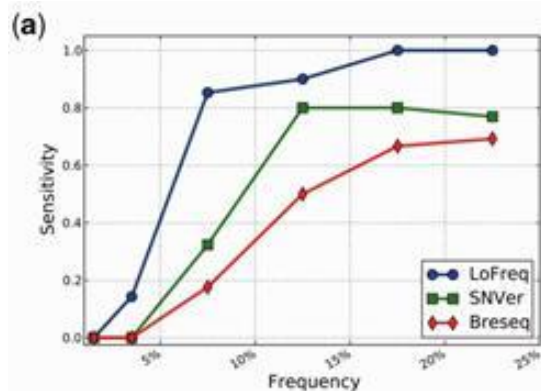
- -q Minimum quality score threshold to count base (Default: 20)
- -t Minimum frequency threshold(0 - 1) to call variants (Default: 0.03)
- -m Minimum read depth to call variants (Default: 0)
- -r Reference file used for alignment. This is used to translate the nucleotide sequences and identify intra host single nucleotide variants
- -g A GFF file in the GFF3 format can be supplied to specify coordinates of open reading frames (ORFs). In absence of GFF file, amino acid translation will not be done.

- Output Options Description

- -p (Required) Prefix for the output tsv variant file

Grubaugh et al. Genome Biology. 2019

LOFREQ



Wilm et al. Nucleic Acids Res. 2012

LOFREQ

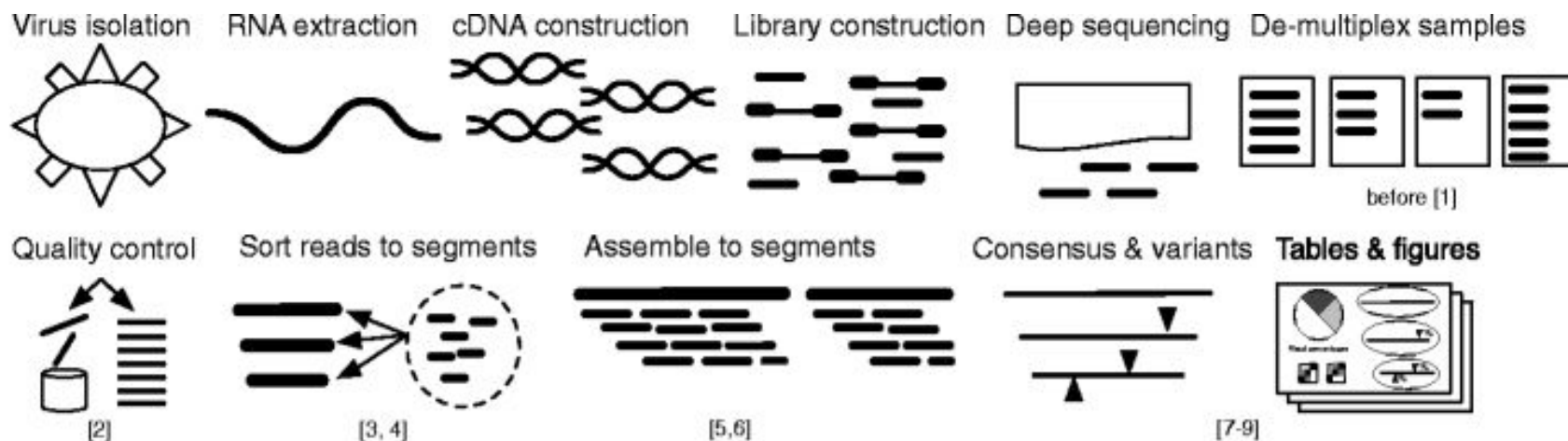
- lofreq viterbi: realignment algorithm
- lofreq call: Warning! Only SNPs are called by default.
- lofreq filter: vcf filtering.

<http://csb5.github.io/lofreq/>

IRMA

IRMA: Iterative Refinement Meta-Assembler

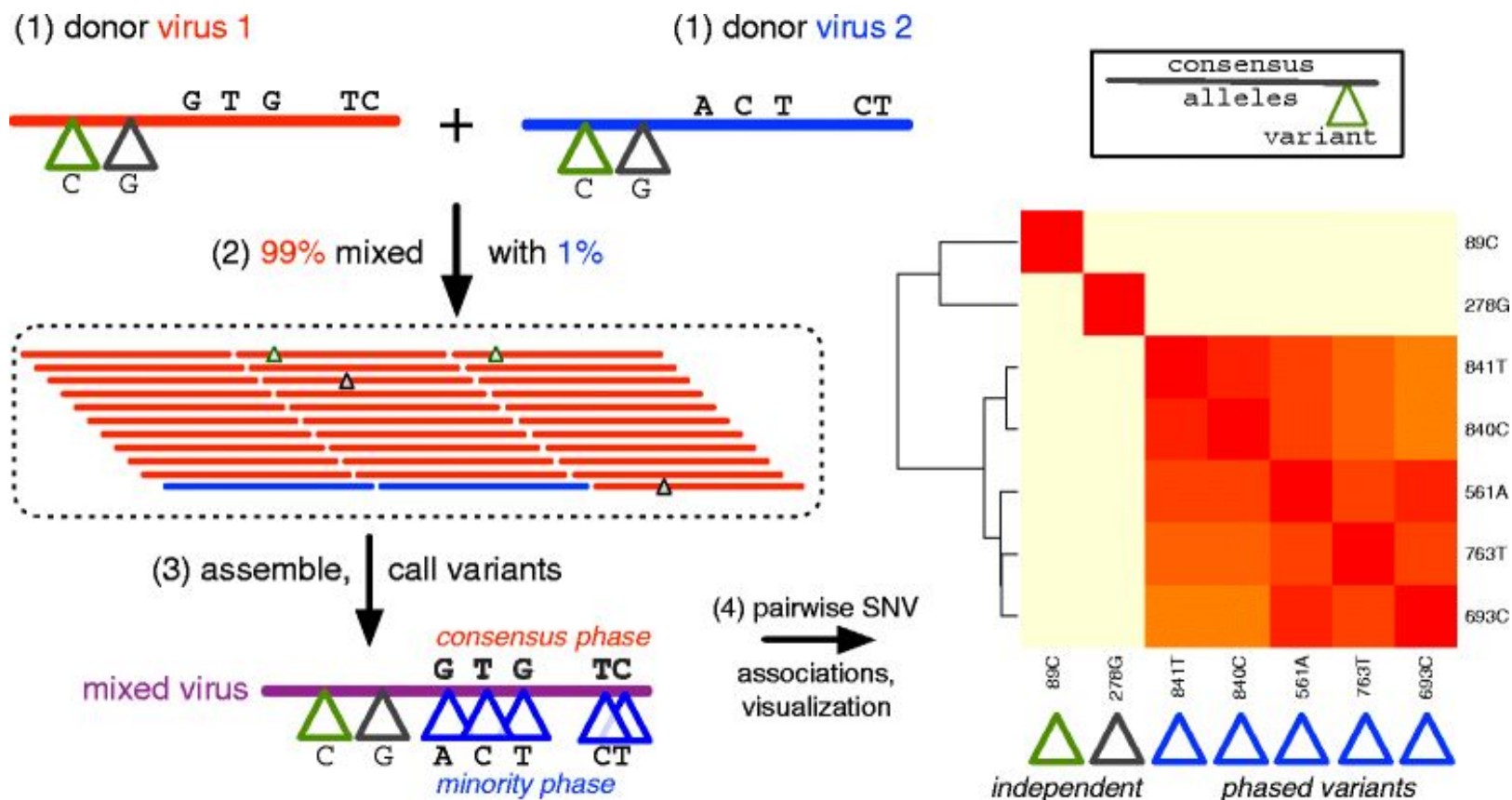
A



Shepard et al BMC Genomics 2016, **17**:708

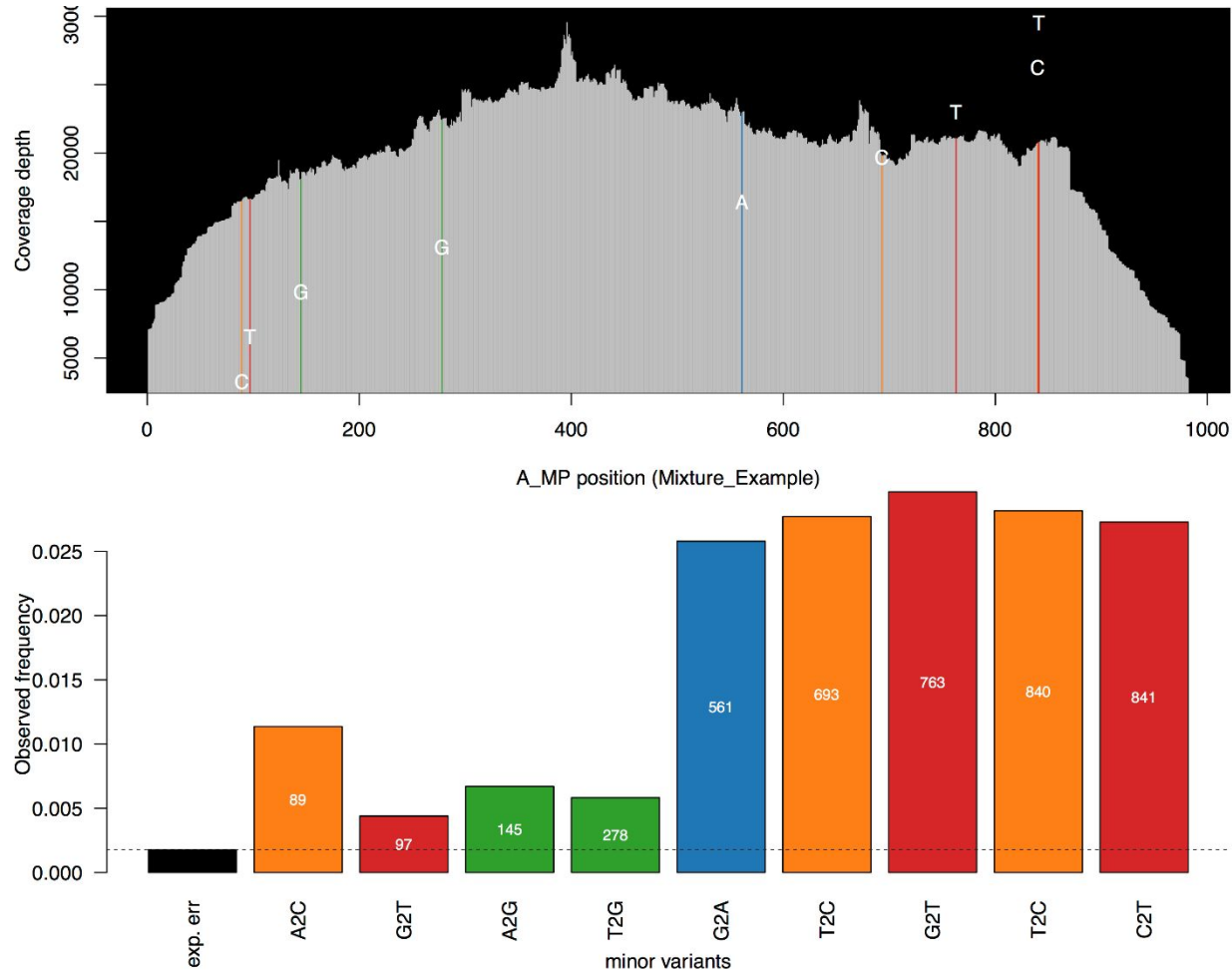
Available for LU, *FLU_AD, EBOLA, & ‡CoV & Sars-cov-2

IRMA



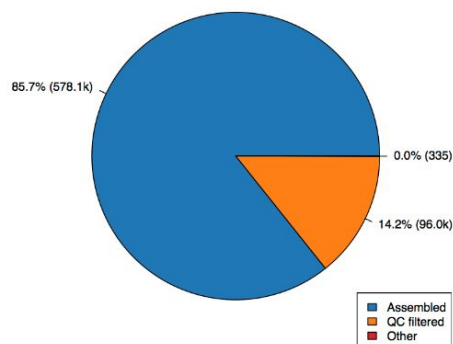
Shepard et al BMC Genomics 2016, 17:708

IRMA

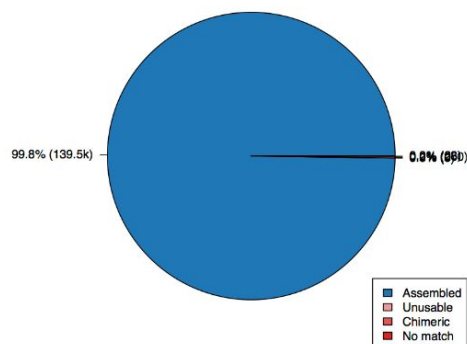


IRMA

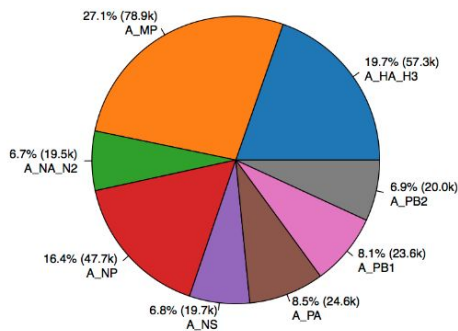
1. Percentages of total reads (R1 + R2)



2. Percentages of all read patterns passing QC



3. Percentages of assembled, merged-pair reads



SAMPLE "Mixture_Example"

READ PROPORTIONS.

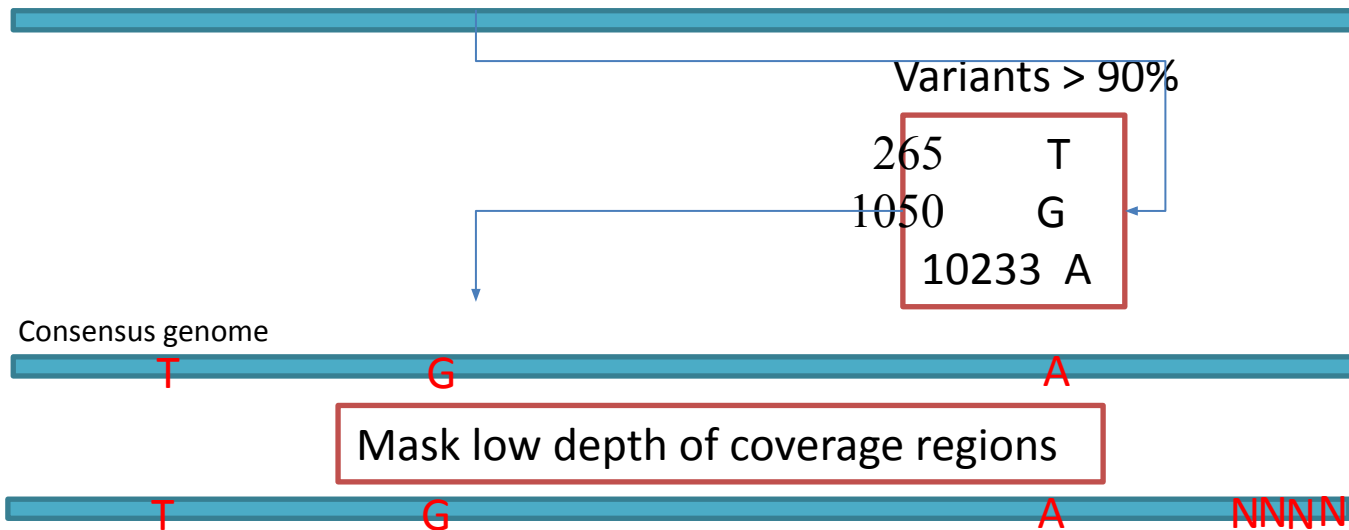
- Percentages of total read counts (R1 & R2)
 - ASSEMBLED: influenza reads in final assemblies.
 - QC FILTERED: didn't pass length/median quality thresholds.
 - OTHER: non-flu and contaminant/poor flu signal.
- Percentages of all read patterns passing QC process
 - Patterns are clustered or non-redundant reads.
 - ASSEMBLED: excellent influenza read patterns.
 - UNUSABLE: poor or contaminant flu patterns.
 - CHIMERIC: flu patterns matching both strands.
 - NO MATCH: non-flu read patterns.
- Percentages of assembled, merged-pair read counts
 - Shows the proportion of gene segments to the genome.
 - Paired-end reads have been merged into a single count unless not applicable: single-end reads have been used.

Consensus genome

Aproximation 1

- Select variants: > 80% allele frequency
- Include variants in reference genome.
- Mask low frequency positions: <10x.

Ref genome



Consensus genome

- **Approximation 2 (ivar)**

- Minimum frequency threshold is the minimum frequency that a base must match to be called as the consensus base at a position. If one base is not enough to match a given frequency, then an ambiguous nucleotide is called at that position

As an example, consider a position with 6As, 3Ts and 1C. The table below shows the consensus nucleotide called at different frequencies.

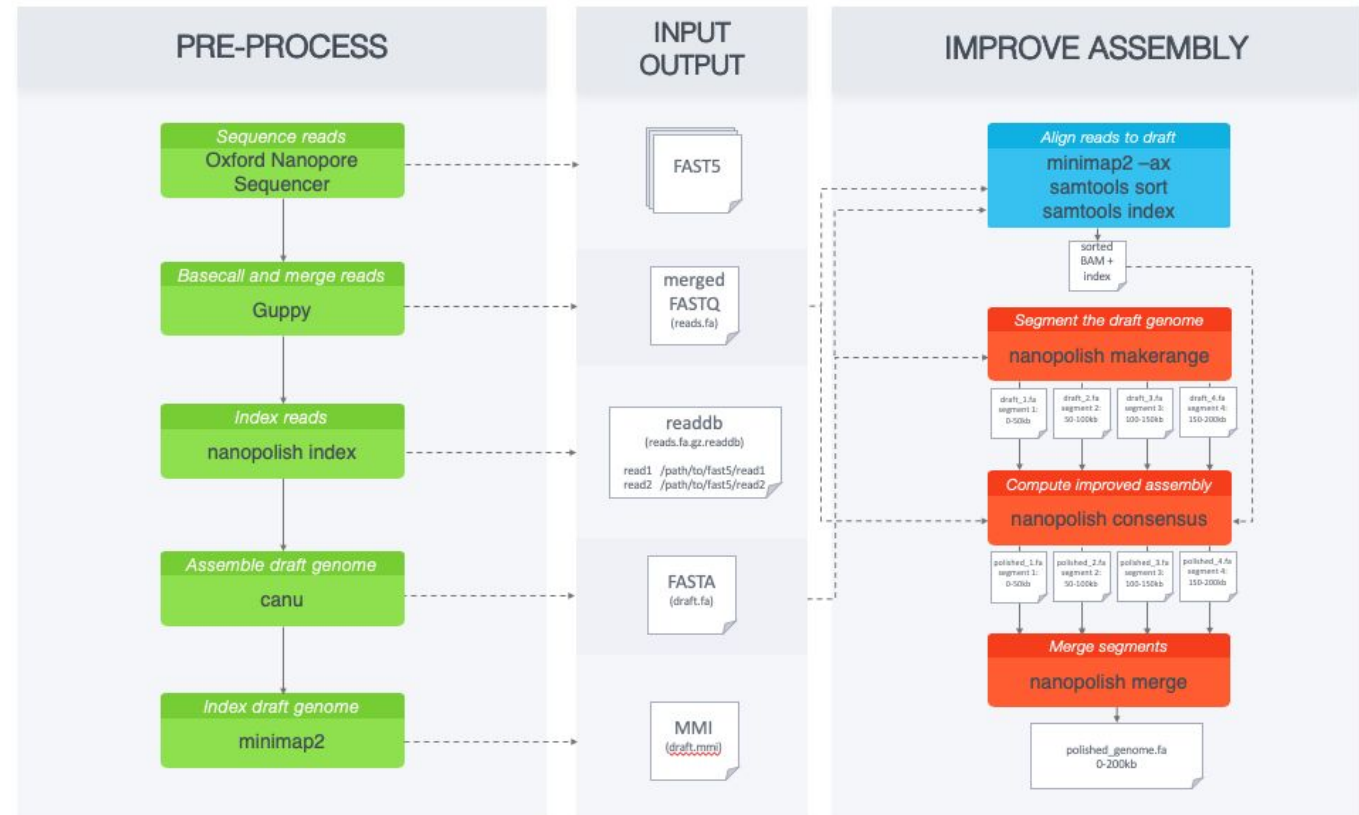
Minimum frequency threshold	Consensus
0	A
0.5	A
0.6	A
0.7	W(A or T)
0.9	W (A or T)
1	H (A or T or C)

Variant Calling in Nanopore

- **Medaka** is a tool to create consensus sequences and variant calls from nanopore sequencing data. This task is performed using neural networks applied a pileup of individual sequencing reads against a draft assembly.
- medaka consensus: Creates a consensus from a draft assembly in fasta format and the nanopore corrected fastq reads.

Variant Calling in Nanopore

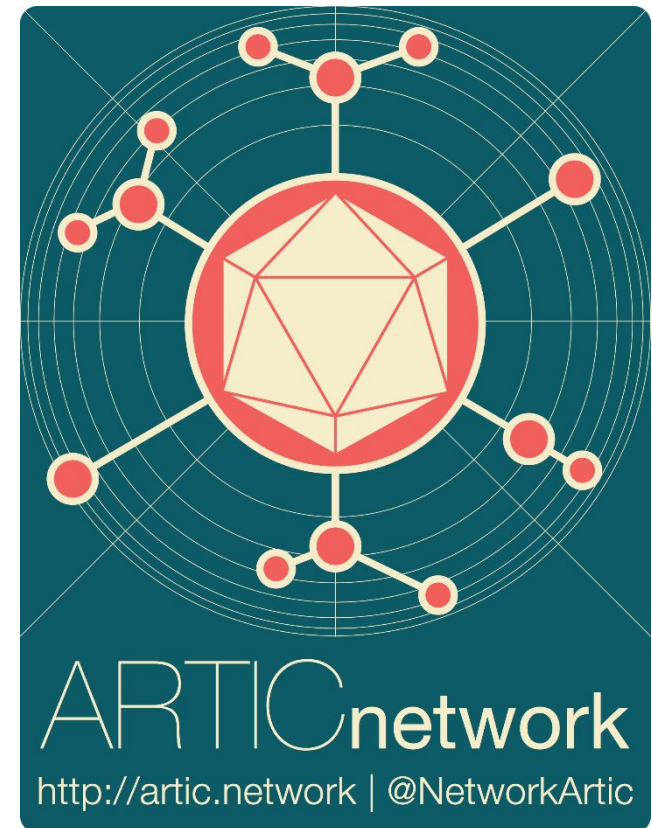
- **Nanopolish** can calculate an improved consensus sequence for a draft genome assembly, detect base modifications, call SNPs and indels with respect to a reference genome and more



Variant Calling in Nanopore

- Artic protocol

Full protocol from wet to drylab.
Includes basecalling, preprocessing,
mapping and consensus generation for
Artic primers.

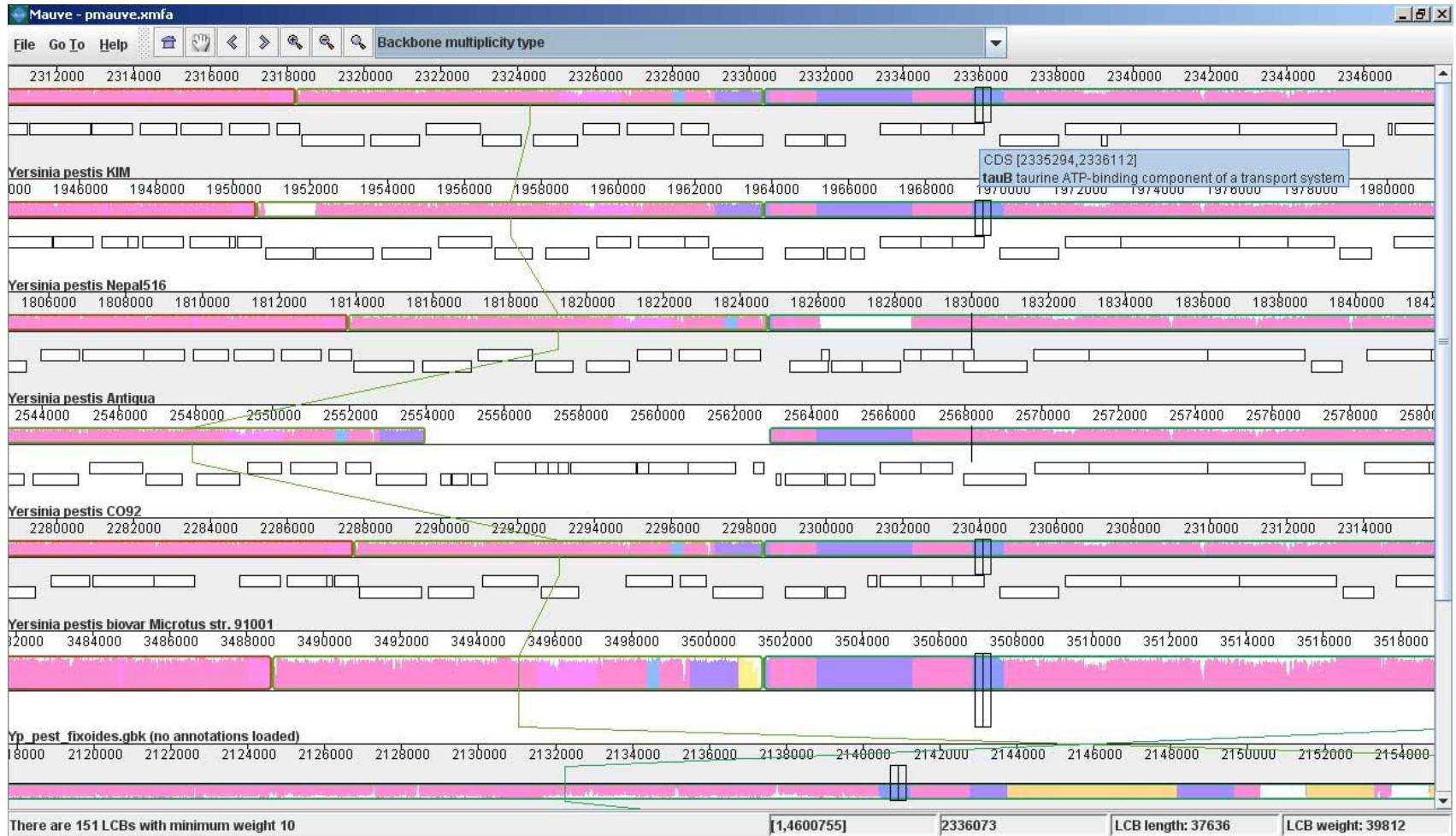


Consensus genome comparison: Mauve

Mauve is a complete genome aligner



Consensus genome comparison: Mauve



Thanks for your attention!
